Random Neighbor Pixel-Block-Based Deep Recurrent Learning for Polarimetric SAR Image Classification

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***Abstract*— Polarimetric synthetic aperture radar (PolSAR) image classification is an important part of SAR data interpretation and provides more intuitive and detailed SAR polarization information. To bridge the PolSAR data and applications, it is necessary to design a comprehensive PolSAR classification framework to achieve satisfactory results. The deep neural network (DNN) appears to be a solution for the classification issue, in which it outperforms the classical supervised classifiers under the condition of sufficient training data. However, the volume of training data will greatly limit the effectiveness of practical applications. In this article, we try to solve the dependence issue on training data in three different ways: recurrent learning, data augmentation, and postprocessing. First, the long short-term memory (LSTM) network is introduced to achieve pixel sequence learning by taking into account the spatial and polarimetric features. Second, the random neighbor pixel-block (RNPB) method is proposed to increase the number of training samples for sequence learning. Third, the conditional random field (CRF) model is employed to further improve the classification accuracy. In the experiments, three sets of PolSAR data are used to evaluate the small sample performance of the proposed classification method. With only 0.5% labeled pixels for training, the proposed RNPB-LSTM-CRF method can approach 99% overall classification accuracy for all the data sets. Compared with the existing methods, the proposed method can achieve state-of-the-art results for PolSAR image classification under the condition of 1% training samples.**

***Index Terms*— Conditional random field (CRF), data augmentation, image classification, long short-term memory (LSTM), polarimetric synthetic aperture radar (PolSAR).**

Manuscript received December 24, 2019; revised June 11, 2020, July 20, 2020, and September 13, 2020; accepted November 5, 2020. Date of publication December 2, 2020; date of current version August 30, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61871413, Grant 61801015, and Grant 61871335; and in part by the Fundamental Research Funds for the Central Universities under Grant XK2020-03. *(Corresponding author: Fan Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2020.3037209

I. Introduction

SYNTHETIC aperture radar (SAR) has been widely used in geographical surveys, climate change research, and other applications because of its high resolution, day-and-night utility, and weather independence [1]. Polarization is highly related to the physical properties of targets, thus reflecting different information provided by the magnitude, frequency, and phase [2]. In the past decades, many polarimetric SAR (PolSAR) systems have been developed and applied in land cover classification [3], soil moisture inversion [4], target recognition [5]–[7], and so on. Due to the orientation, size, dielectric, and moisture characteristics of targets affecting PolSAR data, PolSAR image classification has been a hot topic in remote sensing applications [8], [9].

With the development of machine learning, supervised classification methods have attracted increased attention due to their superior classification performance; these methods include support vector machine (SVM) [10], [11], sparse-representation classification (SRC) [12]–[14], nearest-regularized subspace (NRS) [2], [15], and ensemble learning [16]–[18]. Regarding pixelwise classification, the similarity among pixels is utilized for category discrimination, which not only relies on the single-pixel polarimetric features but also considers pixel-centric neighborhood spatial features [19]. With the advent of deep learning, the deep neural network (DNN) has been widely employed in remote sensing image classification [20]–[23]. To integrate spatial features, a neighborhood preserved DNN (NPDNN) has been proposed for PolSAR feature extraction and classification, which uses network weights to establish the relationships between superpixels and samples [24]. In addition, the convolutional neural network (CNN) has been utilized as an effective method for image feature extraction in PolSAR image classification, e.g., complex-valued CNN, shallow CNN, 3-D CNN, and so on [25]–[28]. Although CNNs can effectively extract the single-pixel polarimetric features, the extraction of spatial features is just a simple method of concatenating or accumulating the features from neighborhood pixels. Therefore, the utilization of spatial features needs to be further improved for deep learning-based classification.

Due to its ability to consider contextual information, the recurrent neural network (RNN) is widely used in SAR imaging, SAR/PolSAR image classification, hyperspectral

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image classification, and so on [29]–[38]. The most significant difference among these existing RNN-based image classification methods mainly lies in how the sequence samples are constructed. For hyperspectral image classification, the spectral features of continuous bands or neighborhood pixels are used to constitute the input samples [30], [32]. For PolSAR image classification, the polarimetric features of continuous observation time or polarimetric orientation angles (POAs) are employed to construct the input sequences [35], [36]. Especially, Geng *et al.* [34] used long short-term memory (LSTM) to extract PolSAR image features, but the coherence and scattering properties of the polarization data were not extracted and utilized. Minh *et al.* [35] used gated recurrent unit (GRU) and LSTM to realize the classification of winter vegetation quality mapping, and they input PolSAR image data at different times into GRU and LSTM to perform classification. Wang *et al.* [36] used different polarization azimuth sequences of the coherence process as learning objects of the convolutional LSTM (ConvLSTM) to achieve advanced feature extraction and classification. Nevertheless, those DNN-based methods usually require a large number of samples to fit the nonlinear model for deep feature extraction. The limited training samples will lead to overfitting of the network model, thus affecting the generalization of the model [39]. The number of training samples is proportional to the classification accuracy: more training samples will lead to better performance. Currently, the augmentation of training samples is an important way to improve the classification accuracy of DNN. For example, Geng *et al.* [40] used the method of superpixel segmentation to divide the Pauli image into many patches and then took the average feature sequence of one patch as an extension of the training set of similar categories; Li *et al.* [41] proposed a pixel-block pair (PBP) method to increase the richness of training samples, and the label of the pixel was generated by voting from the CNN results. Although various augmentation methods are proposed for CNNs, there is seldom any discussion about sequence sample augmentation for RNNs.

According to the aforementioned introduction, an RNN that can extract both polarimetric and spatial features is particularly suitable for PolSAR image classification. In the application process, it is still necessary to further study the recurrent learning strategy and its data augmentation methods to adapt to the characteristics of PolSAR images. Here, we try to represent a PolSAR image pixel with a completed polarimetric feature set for effective high-level feature learning, and we construct the sequence samples with neighborhood pixels for discriminative spatial feature extraction. Dozens of scattering mechanism models are used to construct the polarimetric feature set, e.g., Cloude–Pottier decomposition [42] and Freeman–Dueden decomposition [43]. The LSTM network is utilized to mine deeper features, as well as the relationships of feature sequences between neighborhoods. To alleviate the limited sample problem, the labeled pixels are randomly combined to augment the sequence samples by the random neighbor pixel-block (RNPB) method. In RNPB, the center pixel is original, and the neighbor pixels used to construct the sequence are randomly selected from the labeled pixels to simulate various spatial states. These synthetic samples can both effectively improve the discriminative capability of the LSTM classifier in terms of compact intraclass variation and reduce the misjudgment within a single class block of the PolSAR image. Compared with patch-sorted feature-based learning methods [27], [44], this method can maintain both the authenticity of training samples and the similarity of local neighbor pixels.

In remote sensing image classification, the conditional random field (CRF) is widely used to extend the direct contextual information in both the labeling field and observation field [45]. Regarding PolSAR image classification, CRF has been integrated with several mainstream classification methods, including the maximum likelihood method [46], SVM [47], CNN [48], and deep generative-discriminative network [49] Therefore, the CRF model is applied to the LSTM-based PolSAR classification framework for the fine optimization of labels. Considering the existing deep learning-based PolSAR classification methods, we have made the following contributions.

1) A new recurrent learning strategy for PolSAR image classification is introduced. Based on the LSTM network, it can learn both polarimetric and spatial features from the pixel sequences consisting of neighboring pixels. Each pixel of a sequence is represented by hundreds of polarimetric features.
2) The RNPB data augmentation method is proposed to increase the number of training samples such that the classification accuracy can be improved under the condition of limited samples.
3) The CRF is integrated with LSTM to construct a PolSAR classification framework. Through postprocessing of the probability map yielded by recurrent learning, the intraclass and interclass misjudgments can be further reduced by the energy constraints of the unary and pairwise potentials.

The rest of this article is organized as follows. In Section II, the proposed classification framework is introduced, including the polarimetric target decomposition (TD) models, the LSTM network, the RNPB method, and the CRF model. In Section III, the experimental results and analysis are presented. The conclusion is given in Section IV.

II. Proposed Classification Framework

The proposed classification framework mainly consists of four steps: pixelwise polarimetric feature vector extraction, neighborhood-based sequence sample construction, spatial-polarimetric feature learning-based classification, and classification optimization based on the probability graph model. First, a 3-D feature matrix is constructed for the PolSAR image by integrating various polarimetric features. For each pixel, a high-dimensional feature vector is yielded to comprehensively describe its nature, thereby achieving rich features of limited labeled samples. Next, in both the training and inference stages, each labeled pixel and its neighboring pixels are combined into a sequence sample. In a sequence sample, each pixel is represented by the high-dimensional polarimetric feature vector. For the training process, more

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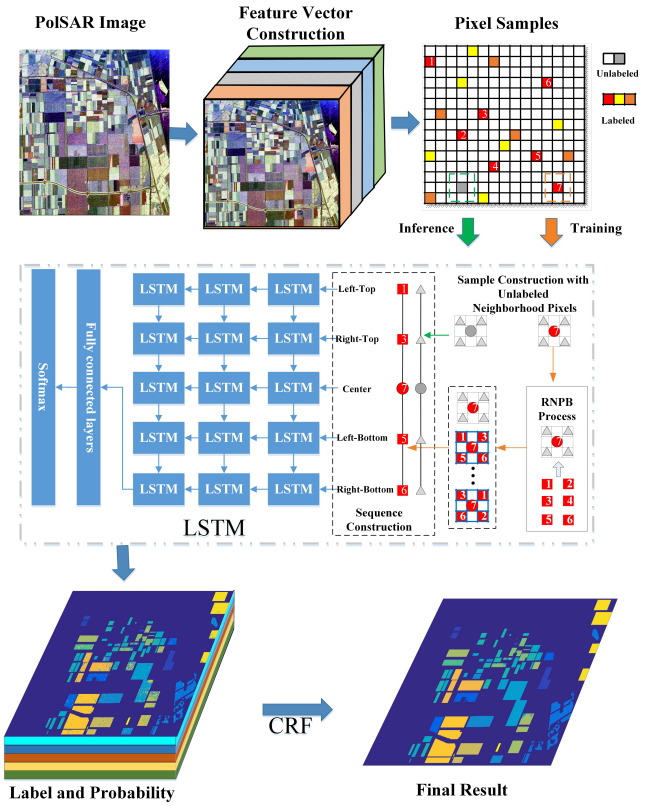


Fig. 1. Proposed PolSAR image classification framework.

synthetic sequence samples of each labeled pixel will be generated by the RNPB method, which is not used in the inference stage. Then, the LSTM network is employed to learn the spatial-polarimetric features of each sequence sample to implement a highly accurate classification. Finally, the calculated labels and probabilities for each pixel are imported into the CRF model to optimize the classification results, as shown in Fig. 1.

A. Pixelwise Polarimetric Feature Vector Extraction

Generally, a PolSAR image contains four channels of information, namely, HH, HV, VH, and VV, which represent four linear orthogonal polarization combinations. Based on the complex data of four channels, three categories of polarimetric feature extraction methods are introduced to improve the discriminative information for each pixel, namely, the TD theorem, matrix elements, and polarimetric descriptors [50], as shown in Table I.

For scattering mechanism interpretation, the TD theorem is usually used to process the complex backscattering matrix to yield the polarimetric features, which can be mainly divided into four categories:

1) features based on the Kennaugh matrix $[K]$ (Huynen, Barnes, and Yang);
2) features based on the scattering model for decomposing the covariance matrix $[C]$ or coherent matrix $[T]$ (Freeman, Freeman and Durden, Yamaguchi, and Neumann);
3) features based on eigenvector and eigenvalue analysis of the covariance matrix $[C]$ or the coherency matrix $[T]$ [Cloude, Holm, van Zyl, Cloude, and Pottier, the multiple-component scattering model (MCSM)];
4) features based on the scattering matrix $[S]$ coherent decomposition method (Krogager, Cameron, and Touzi).

TABLE I
POLARIMETRIC FEATURES EMPLOYED IN THE PROPOSED METHOD

Category	Features (dimension)	Amount
Decomposition components	Barnes(6), Cloude (3), Freeman (2), Freeman-Durden (3), H/A/α (11), Holm (6), Huynen (3), Krogager (3), Vanzyl (3), Yang (7), Yamaguchi (7), Mcsm5 (6), Neuman (3), Tsvm (16)	79
Matrix elements	Coherency Matrix (7), Covariance Matrix (3)	10
Polarimetric descriptors	Span (1),SERD/DERD (1), Polarization Fraction (1), Polarization Asymmetry (1), Shannon Entropy (2), Radar Vegetation Index (1), Radar Vegetation Index (1), Pedestal Height (1), Alpha Approximation (1), Entropy Approximation (2), Scattering Mechanism Entropy (2), Kozlov Anisotropy (1), Lueneburrg Anisotropy (1), Conformity Coefficient (1), Scattering Predominance (1)	18
Total	-	107

For the matrix element features, seven elements of the coherency matrix $[T]$ are extracted: T_{11}, T_{22}, T_{33}, and the amplitude and phase of T_{13} and T_{23}. Another three elements are obtained from the covariance matrix $[C]$: the amplitudes of C_{12}, C_{13}, and C_{23}. In addition, the polarimetric descriptors are developed based on the eigenvalues and eigenvectors of the coherency matrix, such as SPAN, single-bounce eigenvalue relative difference (SERD), double-bounce eigenvalue relative difference (DERD), Shannon entropy, and other features [51]–[55].

To reduce the demand for training samples, 79 features extracted by 14 TD algorithms, 18 polarimetric descriptors, and ten elements of the coherency and covariance matrix are combined into a 107-D polarimetric feature vector for pixel representation [15], as shown in Table I. Then, polarimetric feature extraction is performed on each image pixel to convert the PolSAR image ($M \times N$) into a data cube, denoted as $\mathbf{X} = \{\boldsymbol{x}^{(r,c)}, r \in [1, M], c \in [1, N]\}$ in an $\mathbb{R}^{d\times 1}$ feature space, where (r, c) represents the position of the pixel, M and N are the numbers of rows and columns, respectively, and d is the number of features, namely, 107. Although the polarimetric features come from different perspectives, there are more or less redundancies in the feature vector. The feature selection idea can be adopted to further optimize the vector construction [50], [56].

B. Proposed RNPB Method for Training Sample Augmentation

In deep learning, the number of training samples will directly determine the recognition performance of the network model. RNPB is a sample augmentation method by which

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numerous pixel sequences are generated by randomly selecting the neighbors of the center pixel from the labeled pixels, as shown in Fig. 2. For simplification, only four neighbor pixels are combined to construct a pixel block, which can be further expanded to a sequence for the training and inference process of the LSTM networks.

It is assumed that a PolSAR image has K labeled pixels, as $\mathbf{X}_L = \{\boldsymbol{x}_l^{(r,c)}, l \in [1, K]\}$ in an $\mathbb{R}^{d\times 1}$ feature space and class labels $y_l \in \{1, 2, \ldots, L\}$, where L is the number of categories. Normally, each labeled pixel is concatenated with its neighboring pixels to construct a sequence sample as follows:

$$S = \left\{\boldsymbol{x}_l^{(r,c)}, \boldsymbol{x}^{(r-1,c-1)}, \boldsymbol{x}^{(r-1,c+1)}, \boldsymbol{x}^{(r+1,c-1)}, \boldsymbol{x}^{(r+1,c+1)}\right\}. \tag{1}$$

In this way, the number of sequence samples will be the same as the number of labeled pixels. However, the labeled pixels can be utilized to generate various combined sequences that simulate the spatial distribution of land cover, such as the interior or the edge of a single class block. It is obvious that these synthetic RNPB samples will enhance the performance of intraclass discrimination and interclass discrimination, and they are represented as follows:

$$\begin{cases} S_{\text{inter}} = \left\{\boldsymbol{x}_l^{(r,c)}, \boldsymbol{x}_{\text{rndl}(1)}, \boldsymbol{x}_{\text{rndl}(2)}, \boldsymbol{x}_{\text{rndl}(3)}, \boldsymbol{x}_{\text{rndl}(4)}\right\} \\ S_{\text{edge}} = \left\{\boldsymbol{x}_l^{(r,c)}, \boldsymbol{x}_{\text{rnd}(1)}, \boldsymbol{x}_{\text{rnd}(2)}, \boldsymbol{x}_{\text{rnd}(3)}, \boldsymbol{x}_{\text{rnd}(4)}\right\} \end{cases} \tag{2}$$

where rndl$(\cdot)$ represents the random selection from the labeled pixels of the l class, and rnd$(\cdot)$ represents the random selection from the labeled pixels.

As shown in Fig. 2, two categories of RNPB samples can be constructed for the improvement of different misclassifications. To synthesize the homogeneous samples of the red category, the neighbors of the center pixel should be randomly selected from the red class to represent the homogeneous distribution. The number of augmented samples S_{inter} is equal to $P_{K_l}^4$, where P indicates the permutation operation in mathematics and K_l represents the number of labeled pixels in class l. Similarly, numerous heterogeneous synthetic samples will improve the recognition performance of the edge pixels in the class block. For the heterogeneous samples of the red class, the neighbors of the center pixel can be randomly selected from all of the classes. Thus, the number of synthetic samples S_{edge} will be P_K^4. For the intraclass discrimination, the RNPB samples are increased by $(K_l - 1)!/(K_l - 5)!$ times, which can guarantee the sufficient training sequences for spatial-polarimetric feature recurrent learning.

C. Spatial-Polarimetric Feature Learning With the LSTM Network

As a special form of RNN, the LSTM network [57] has been widely employed to learn the contextual features from sequence samples in various fields. Based on the polarimetric feature vector, the LSTM network can be employed to further extract the spatial features among neighborhood pixels. As illustrated in Fig. 1, the LSTM network consists of three LSTM layers, one fully connected (FC) layer, and one softmax

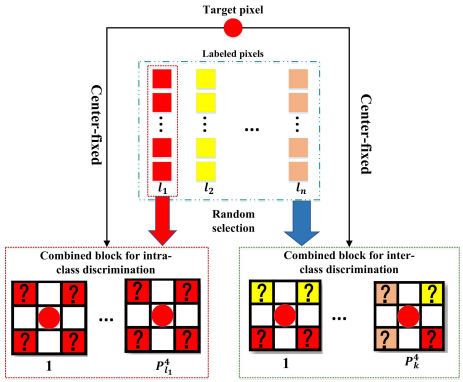


Fig. 2. RNPB sample augmentation.

layer. For each LSTM layer, the number of LSTM units is the same as the length of the sequence sample, while the size of LSTM units should be adapted to the feature dimension. Considering that each RNPB sample is composed of five neighboring pixels and each pixel is represented by 107-D features, the first LSTM layer should have five LSTM units, each of which includes 128 input connections. Assuming that there are 11 classes in any given PolSAR image, the output shapes of the three LSTM layers and FC layer are $(5, 128)$, $(5, 64)$, (32), and 11, respectively.

The basic principle of the LSTM cell is shown in Fig. 3. After the extraction of the polarimetric feature vector for each pixel, the RNPB samples are denoted as $S_{rnpb} = \{\boldsymbol{x}_t, t \in [1, 5]\}$ in $\mathbb{R}^{d\times 1}$ feature space. For the pixel $\boldsymbol{x}_t$, the mapping function of the LSTM cell is specified by

$$\begin{cases} \text{Input Gate}: & i_t = \sigma(W_{ix} \cdot x_t + W_{ih} \cdot h_{t-1} + b_i) \\ \text{Forget Gate}: & f_t = \sigma(W_{fx} \cdot x_t + W_{fh} \cdot h_{t-1} + b_f) \\ \text{Output Gate}: & o_t = \sigma(W_{ox} \cdot x_t + W_{oh} \cdot h_{t-1} + b_o) \\ \text{Input Node}: & I_t = \tanh(W_{Ix} \cdot x_t + W_{Ih} \cdot h_{t-1} + b_I) \\ \text{Cell State}: & C_t = f_t * C_{t-1} + i_t * I_t, \\ \text{Cell Output}: & h_t = o_t * \tanh(C_t) \end{cases} \tag{3}$$

where t represents the spatial dimension (the pixel position in the neighboring pixel sequence), and σ is the logistic sigmoid function. W and b, respectively, represent weight matrices and bias vectors. After sequence feature extraction by the LSTM layers, the softmax layer is employed for classification, which is defined as

$$P_t(l) = \frac{\exp\{W_{lh}h_t + b_l\}}{\sum_{j=1}^{L}\exp\{W_{jh}h_t + b_j\}}. \tag{4}$$

In each LSTM layer, the cell units are recurrently concatenated. Due to this kind of connection, the contextual information can be transferred freely, as shown in the top connecting branch of the LSTM unit. Moreover, the most special design is the cell state, which is employed to store the long-term history information C_{t-1} and short-term input information $\boldsymbol{x}_t$. Through three control gates, each LSTM unit will learn to memorize the most discriminative parts among the history features and the input features. Especially, the cell

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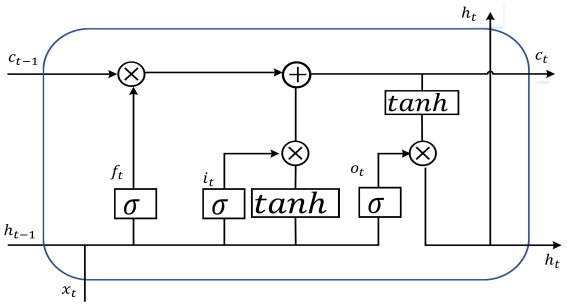


Fig. 3. Basic structure of the LSTM unit.

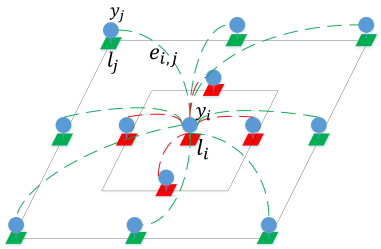


Fig. 4. Illustration of the energy in the CRF model.

state could be expanded into (5), i.e.,

$$C_t = f_t * C_{t-1} + i_t * \tanh(W_{Ix} \cdot x_t + W_{Ih} \cdot o_{t-1} * \tanh(C_{t-1}) + b_I) \quad (5)$$

from which it can be determined that the contextual features are formed by considering both long-term $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{t-1}\}$ and short-term $\boldsymbol{x}_t$ features. In other words, the long- and short-term features, respectively, represent the spatial dimension and the polarimetric dimension. Therefore, the spatial-polarimetric features will be extracted with the training of LSTM networks.

D. CRF Probability Map Optimization

Since the image includes the spatial contextual information, the CRF can be used to refine the classification results. By considering the probability distribution, pixel position, and pixel intensity, the relationships between each pixel and neighboring pixels are built with an FC CRF model [58], which is shown in Fig. 4.

In Fig. 4, $e_{i,j}$ represents the energy relationship between the center node y_i and its neighborhood node y_j. The green dotted lines that connect the nodes farther away from y_i illustrate the weak relations, indicating that the influence between the two nodes is relatively small. The red lines that connect the nodes closer to y_i indicate strong relations between the two nodes. Therefore, the energy function between the central node and the neighborhood is expressed as follows:

$$E(l) = \sum_i \xi_u(l_i) + \sum_{i<j} \xi_p(l_i, l_j) \quad (6)$$

where $\sum_i \xi_u(l_i)$ carries probability information, which is calculated by (4). l_i is the label of pixel i. $\xi_p(l_i, l_j)$ is the pairwise potential function, which is used to further constrain the labeling based on the spatial relationships of pixel position and intensity, whose form is given by

$$\xi_p(l_i, l_j) = u(l_i, l_j) \cdot w \cdot \exp\left\{-\frac{|l_i - l_j|^2}{2\theta^2} - \frac{|I_i - I_j|^2}{2\theta^2}\right\} + u(l_i, l_j) \cdot \exp\left\{-\frac{|l_i - l_j|^2}{2}\right\} \quad (7)$$

where $u(\cdot)$ is a label compatibility function, $u(l_i, l_j) = [l_i \neq l_j]$, w is the linear combination weight, I is the pixel intensity, and θ is used to control the degrees of nearness and similarity. The parameter tuning issue can be addressed by the simple grid search method [59].

The detailed steps of the proposed method are shown in Algorithm 1.

Algorithm 1 Proposed Method for PolSAR Image Classification

Input: polarimetric scattering matrix $[S]$ of PolSAR image, the labels of the training pixel samples.
Output: the labels L for PolSAR image.

```
Refined Lee speckle denoising;
Pixelwise feature vector construction from the decomposition components, matrix elements and polarimetric descriptors;
Construct sequence sample S with neighborhood pixels;
for each labeled pixel do
    yield the synthetic sequence samples S_edge and S_inter with the RNPB method;
end for
while not meet the termination criterion do
    Solve LSTM learning-subproblem to update the network parameters;
end while
Generate the probability map P of PolSAR image;
while not reach the preset iterations do
    update the probability map P with CRF model;
end while
Get the labels L according to the final probability map P.
```

III. Experimental Results and Analysis

Three experiments with different airborne PolSAR data are designed to investigate the effectiveness of the proposed method. The employed CNN and LSTM networks are implemented under the Keras architecture, and the basic configuration of the platform is shown in Table II. In the experiments, the training sequence samples consist of two parts: 1) original samples: a labeled pixel and its four neighborhood pixels are selected to construct the sequence sample, such that one labeled pixel will generate only one sample and 2) synthetic samples: a labeled pixel and four randomly selected pixels from the labeled pixels of the same class are employed to construct many samples. For the inference stage, an unlabeled pixel and its four neighborhood pixels are selected to construct the sequence sample for category recognition.

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TABLE II
CONFIGURATION OF EXPERIMENTAL PLATFORM

Programming language	Python 3.5
Deep learning API	Keras 2.1.6
CPU	Intel core i7-7700k
Memory	16GB
GPU	GeForce GTX 1060

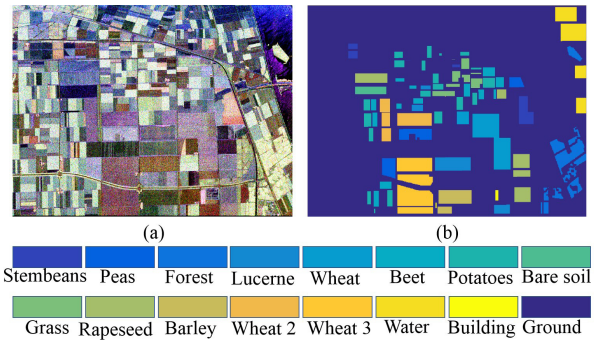


Fig. 5. Flevoland I data set and its color code. (a) Pauli RGB map. (b) Ground-truth map.

TABLE III
EXPERIMENTAL DATA OF FLEVOLAND I

Class	Name	Train	Expand	Test
1	Stem beans	31	124	6103
2	Peas	46	184	9111
3	Forest	75	300	14944
4	Lucerne	48	192	9477
5	Wheat	87	348	17283
6	Beat	51	204	10050
7	Photos	77	308	15292
8	Bare soil	16	64	3078
9	Grass	32	128	6269
10	Rapeseed	64	256	12690
11	Barley	36	144	7156
12	Wheat 2	53	212	10591
13	Wheat 3	107	428	21300
14	Water	68	272	13476
15	Building	6	24	476
Total	-	797	3188	157296

A. Experimental Data

1) AIRSAR Data in Flevoland I: The first quad-PolSAR data set includes L-band data acquired by the NASA/JPL AIRSAR system in Flevoland, The Netherlands, in August 1989. The incidence angle is approximately 20° at the near range and 44° at the far range. The size of these data is 750×1024 pixels. The ground-truth data include 157 296 pixels, including 15 different land cover types, as shown in Fig. 5. In the experiment, 0.5% of the labeled samples are used as the training samples, which are expanded by four times using the RNPB method. The experimental information is shown in Table III.

2) AIRSAR Data in Flevoland II: The second quad-PolSAR data set was acquired in the same area by the AIRSAR system in June 1991. The incidence angles are approximately 20°–44°, and there are 13 different land cover types marked in the ground truth. The size of these data is 1279×1024 pixels, among which 122 928 pixels are labeled. The Pauli image and its ground-truth image are shown in Fig. 6. In the experiment, the specific sample configuration is shown in Table IV.

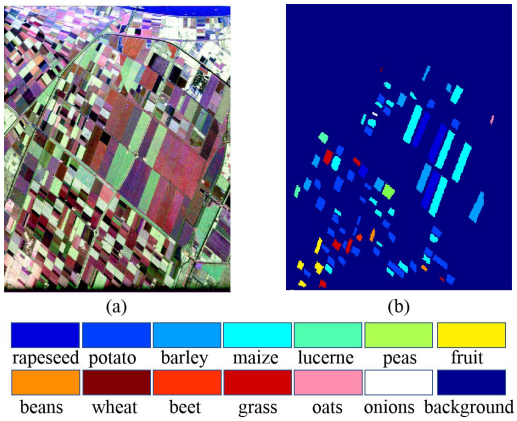


Fig. 6. Flevoland II data set and its color code. (a) Pauli RGB map. (b) Ground-truth map.

TABLE IV
EXPERIMENTAL DATA OF FLEVOLAND II

Class	Name	Train	Expand	Test
1	Rapeseed	89	356	17621
2	Potato	165	660	32854
3	Barley	83	332	16475
4	Maize	201	804	40019
5	Lucerne	12	48	2336
6	Peas	13	52	2409
7	Fruit	21	84	4156
8	Beans	7	28	1258
9	Wheat	6	4	120
10	Beet	6	12	559
11	Grass	30	120	5800
12	Oats	6	8	360
13	Onions	6	8	247
Total	-	645	2516	122928

TABLE V
EXPERIMENTAL DATA OF GF-3

Class	Name	Train	Expand	Test
1	Bare	24	96	23038
2	Forest	40	160	39099
3	Cole	50	200	49588
4	Wheat	36	144	35121
5	Grass	11	44	10214
6	Water	6	24	5382
7	Sand	11	44	10213
8	Wetland	6	4	893
Total	-	184	716	173548

3) GF-3 Data in Hulunbeir: The third experimental quad-PolSAR data set was acquired in Hulunbeir by the GF-3 system of China, including eight different land cover types. The size of these data is 1147×1256 pixels, and 173 548 pixels are labeled. The Pauli image and its ground-truth image are shown in Fig. 7. In the experiment, 0.1% of samples are selected as the training samples. Based on these labeled samples, four times expanded samples are generated by the RNPB method. The experimental data configuration is shown in Table V.

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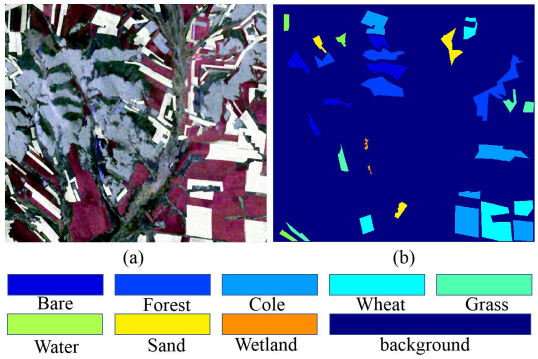


Fig. 7. GF-3 data set and its color code. (a) Pauli RGB map. (b) Ground-truth map.

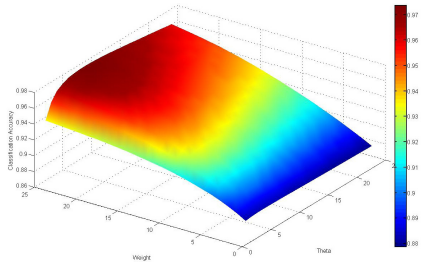


Fig. 8. Grid search results for CRF hyperparameters: w and θ.

B. Parameter Settings

For the proposed method, there are a few parameters that should be determined in advance, including the model hyperparameters and training parameters. Therefore, a series of experiments with 0.5% labeled pixels of the Flevoland data set is used to find the corresponding settings.

First, considering the pixel number of a sequence sample and the 107-D input features for each pixel, each LSTM layer has five LSTM units, and the input size of the LSTM unit in the first layer is set to 128. Because the number of categories is 15, the input size of the LSTM unit in the last layer should be greater than 15. To set the appropriate network structure, the experiments with two and three LSTM layers are carried out. The results show that the overall accuracy (OA) of the three-layer structure is approximately 0.5% higher than that of the two-layer structure. Therefore, the sizes of the first, second, and third LSTM layers are 128, 64, and 32, respectively. According to the RNPB method, the size of each class of training samples should be greater than five. Therefore, if the number of training samples in a certain category is less than five, the number of these training samples is set to six in the experiments. There is a tradeoff between efficiency and accuracy with respect to the number of augmented samples. Therefore, the multiples of synthetic samples to original training samples are set from 0 to 10 to evaluate the classification performance. When the multiple exceeds 4, the OA essentially remains at approximately the same level. Therefore, 0.5% of original and 2% of synthetic sequence samples are used as training samples in the experiments.

Second, as observed in (7), there are two hyperparameters: w and θ. Through the grid search, the relation between the parameter and classification accuracy is shown in Fig. 8, where the optimal range of w is approximately 15–25 and that of θ is approximately 5–15. Thus, in the experiments, w is set to 20, and θ is set to 10. In addition, there is an undetermined parameter in the inference stage, namely, the number of iterations for successively optimizing the energy function. Through experimental searching, the number of iterations is set to 5 by considering accuracy and efficiency.

Furthermore, the training settings are consequently described, including epoch, optimizer, and batch size. The number of epochs is set to 150 in the experiments. In addition, by considering the hardware environment shown in Table II, the batch size is set to 16. Moreover, Adam is employed as an optimizer to realize the adaptive adjustment of the learning rate, and its initial value is set to 0.001 [60].

C. Classification Accuracy

To evaluate the performance of the proposed method, five methods are employed for accuracy comparison. The CNN-based methods are derived from the 1-D CNN network, and its input features are composed of concatenated polarimetric features from the neighborhood pixels. The spatial-polarimetric features can be learned by a CNN, which will contain the same amount of input feature information as LSTM. These comparison methods are described as follows.

1) CNN-9 indicates that the number of neighborhood pixels is 9.
2) CNN indicates that the number of neighborhood pixels is 5.
3) RNPB-CNN means that the training samples in the CNN method are augmented by the RNPB method.
4) LSTM represents the standard LSTM network part of the proposed method.
5) RNPB-LSTM means that the training samples are increased by the RNPB method on the basis of LSTM.

1) Results and Analysis of the Flevoland I Data: First, the AIRSAR data set I in Flevoland is used to evaluate the effectiveness of the LSTM method and its combination with the RNPB method. Then, it is further integrated with the CRF model to acquire the final classification results, which are shown in Fig. 9 and Table VI.

Especially, Fig. 9(a)–(c) shows the results of CNN-based methods, which concatenates the neighborhood pixel features as the spatial-polarimetric feature representation of the center pixel. It can be found that the RNPB method can improve the accuracy of the CNN method by more than 1.3% in this experiment.

In contrast, Fig. 9(d)–(f) shows the results of LSTM-based methods, in which the sequence samples are concatenated by five neighboring pixel features. The accuracy of LSTM-based classification is 86.37%, while the accuracy can be improved by 2.37% with the RNPB method. In more detail, the advantage of the RNPB method is shown in Fig. 10. The classification accuracies of Lucerne (blue) and Wheat-3 (orange) are 89.94% and 91.14%, respectively. After being augmented by the RNPB method, the misjudgments of the blue block and

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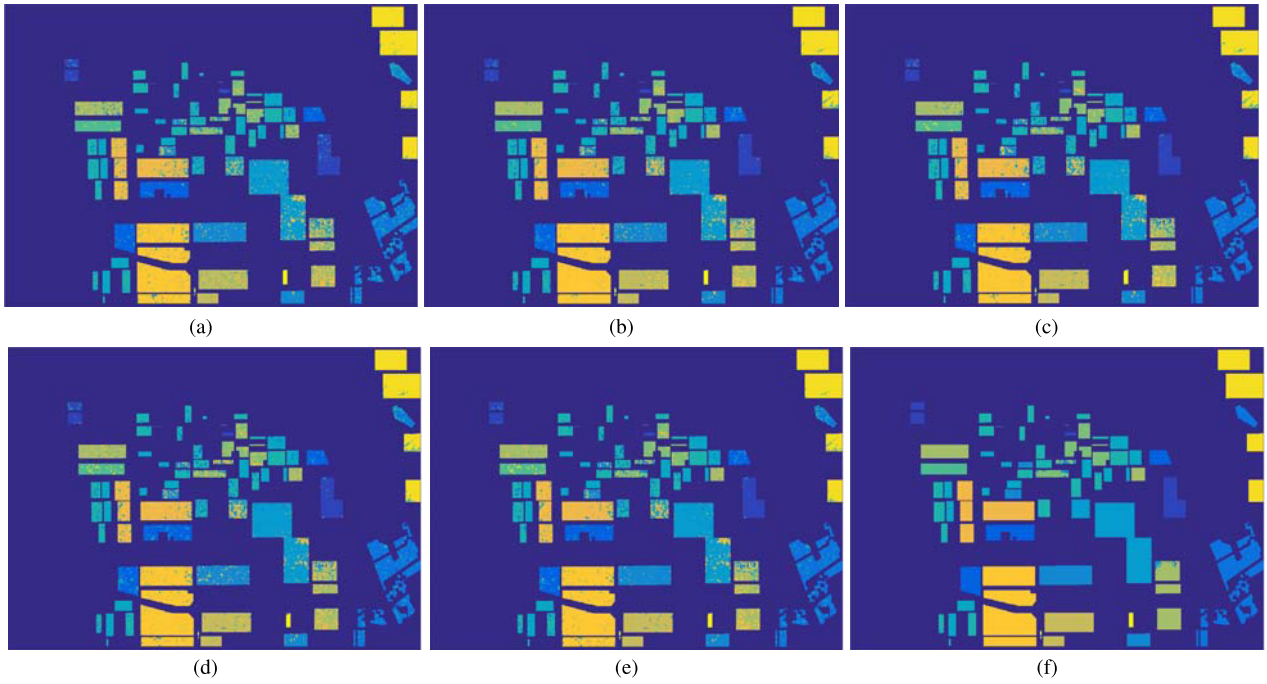


Fig. 9. Classification maps of different methods on Flevoland I data set. (a) CNN-9, OA = 83.9%. (b) CNN, OA = 86.02%. (c) RNPB-CNN, OA = 87.4%. (d) LSTM, OA = 86.37%. (e) RNPB-LSTM, OA = 88.74%. (f) RNPB-LSTM-CRF, OA = 99.24%.

TABLE VI
CLASSIFICATION ACCURACY (%) COMPARISON ON THE FLEVOLAND I DATA SET

Class	CNN-9	CNN	RNPB -CNN	LSTM	RNPB -LSTM	Proposed method
1	89.6	95.64	96.05	95.33	96.18	**99.66**
2	94.61	89.16	93	88.31	92.15	**99.95**
3	77.76	88.76	90.14	88.36	89.65	**95.24**
4	90.91	89.8	92.2	89.94	94.92	**99.73**
5	82.42	82.79	84.92	86.08	86.42	**98.17**
6	88.21	94.73	92.29	91.93	96.2	**98.45**
7	90.46	83.98	84.27	86.55	85.48	**99.58**
8	96.04	95.81	93.18	89.54	89.05	**100**
9	69.8	61.17	68.58	60.58	72.61	**99.66**
10	63	68.79	71.75	68.18	72.47	**100**
11	73.62	80.86	91.45	88.08	94.77	**99.87**
12	75.04	81.11	80.28	82.84	82.26	**99.37**
13	87.46	92.92	92.16	91.14	93.1	**99.97**
14	97.27	94.89	94.41	95.24	96.57	**100**
15	90.34	92.23	88.66	83.19	90.76	**100**
OA	83.9	86.02	87.4	86.37	88.74	**99.24**
AA	84.44	86.17	87.55	85.69	88.84	**99.31**
Kappa	0.82	0.85	0.86	0.85	0.88	**0.99**

the orange block are reduced. Accordingly, the classification accuracies are increased by 4.98% and 1.96%. In Fig. 10 (middle), the enlarged image results demonstrate that the RNPB method can effectively reduce the intraclass noises.

From the results in Table VI, it is obvious that the RNPB method can easily improve the average accuracy (AA) regardless of the RNPB-CNN method or the RNPB-LSTM method. Otherwise, it should be noted that their AA values are higher than the OA, which shows that the RNPB method has greatly improved the balance of training samples. In more detail, the classification accuracies of class 9, class 11, and class 15 have been significantly improved by utilizing the RNPB



Fig. 10. Comparison of the classification map details on the Flevoland I data set. (a) LSTM method. (b) RNPB-LSTM method.

method. Compared with other categories, the training samples of class 9, class 11, and class 15 are 32, 36, and 6, respectively, which are slightly insufficient for the complete training. However, after using the RNPB method, the classification accuracies have been increased by 12.03%, 6.69%, and 7.69%, respectively.

2) Results and Analysis of the Flevoland II Data: The experimental parameters of the Flevoland II data are consistent with those of the previous experiments, and the corresponding results are shown in Fig. 11. By comparing Fig. 11(b) and (c) with Fig. 11(d) and (e), it can be found that the accuracy has been improved by more than 1% with the augmentation of the RNPB method.

The final accuracy can reach 98.94% with the CRF model. The specific results are shown in Table VII. For class 8 and class 10, there are only seven and six pixels for training, respectively, and the accuracies have been improved by more than 10% compared with that of the baseline CNN or LSTM. This result also illustrates the advantage of the proposed method on small data sets. In contrast with the previous experiments, the imbalance of the samples is increased in

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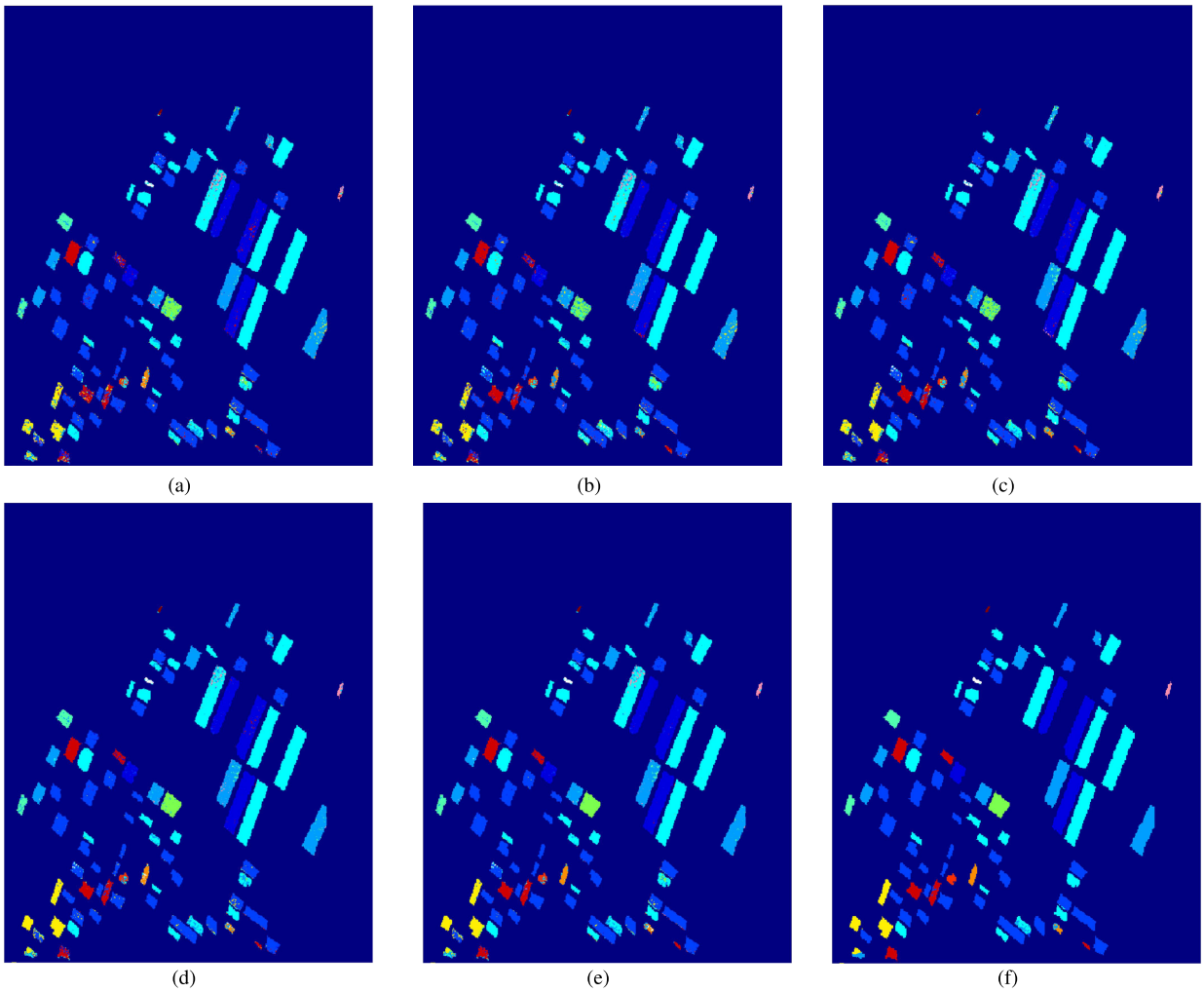


Fig. 11. Classification maps of different methods on the Flevoland II data set. (a) CNN-9, OA = 91.08%. (b) CNN, OA = 90.92%. (c) RNPB-CNN, OA = 91.85%. (d) LSTM, OA = 93.85%. (e) RNPB-LSTM, OA = 95.30%. (f) RNPB-LSTM-CRF, OA = 98.94%.

TABLE VII

CLASSIFICATION ACCURACY (%) COMPARISON ON THE FLEVOLAND II DATA SET

Class	CNN-9	CNN	RNPB -CNN	LSTM	RNPB -LSTM	Proposed method
1	96.16	95.14	97.24	97.96	99.47	**99.99**
2	94.33	93.86	94.66	95.14	97.62	**99.21**
3	92.49	92.17	91.31	95.17	93.93	**94.85**
4	93.41	93.6	93.08	93.64	94.96	**99.63**
5	84.03	88.66	93.58	86.6	95.8	**99.91**
6	81.78	70.2	80.53	95.39	95.72	**100**
7	64.45	71.7	69.13	86.6	84.7	**99.79**
8	51.83	51.11	51.11	63.59	59.38	**100**
9	84.17	86.67	89.17	87.5	90	**100**
10	47.58	55.1	66.01	61.72	83.54	**100**
11	77.86	75.84	86.03	90.48	92.86	**99.81**
12	68.89	83.33	83.61	78.33	87.5	**100**
13	76.92	77.73	63.97	83.4	84.62	**100**
OA	91.08	90.92	91.85	93.85	95.3	**98.94**
AA	77.99	79.62	81.49	85.81	89.24	**99.48**
kappa	0.88	0.88	0.89	0.92	0.94	**0.98**

the configuration of experiments. The category with the most samples includes 201 training pixel samples and 40 019 test pixel samples, while the category with the least samples contains only six training pixel samples and 120 test pixel samples. From the results in Table VII, it is obvious that the LSTM model outperforms the CNN model in terms of all three indices. Although there are only six training samples of categories 9, 10, 12, and 13, the accuracies are improved by 2.5%, 21.82%, 9.17%, and 1.22%, respectively, after RNPB augmentation. Finally, after CRF segmentation, the OA, AA, and kappa reach 98.94%, 99.48%, and 0.9864, respectively.

3) Results and Analysis of the GF-3 Data: The classification results of the GF-3 data are shown in Fig. 12. Comparing Fig. 12(b) and (c) with Fig. 12(d) and (e), it is clear that the accuracy has been improved by more than 1% with the help of the RNPB method. Finally, the optimal accuracy is 99.75%. The specific results are shown in Table VIII. For class 6 and class 8 with only six training pixels, the accuracies have been improved by more than 15% compared with that of the baseline CNN or LSTM classification. This result also illustrates the advantage of the proposed approach on small data sets.

From the results of the above three experiments, three conclusions can be drawn: 1) the LSTM network outperforms the 1-D-CNN in learning the spatial-polarimetric features; 2) the RNPB method can improve the spatial feature

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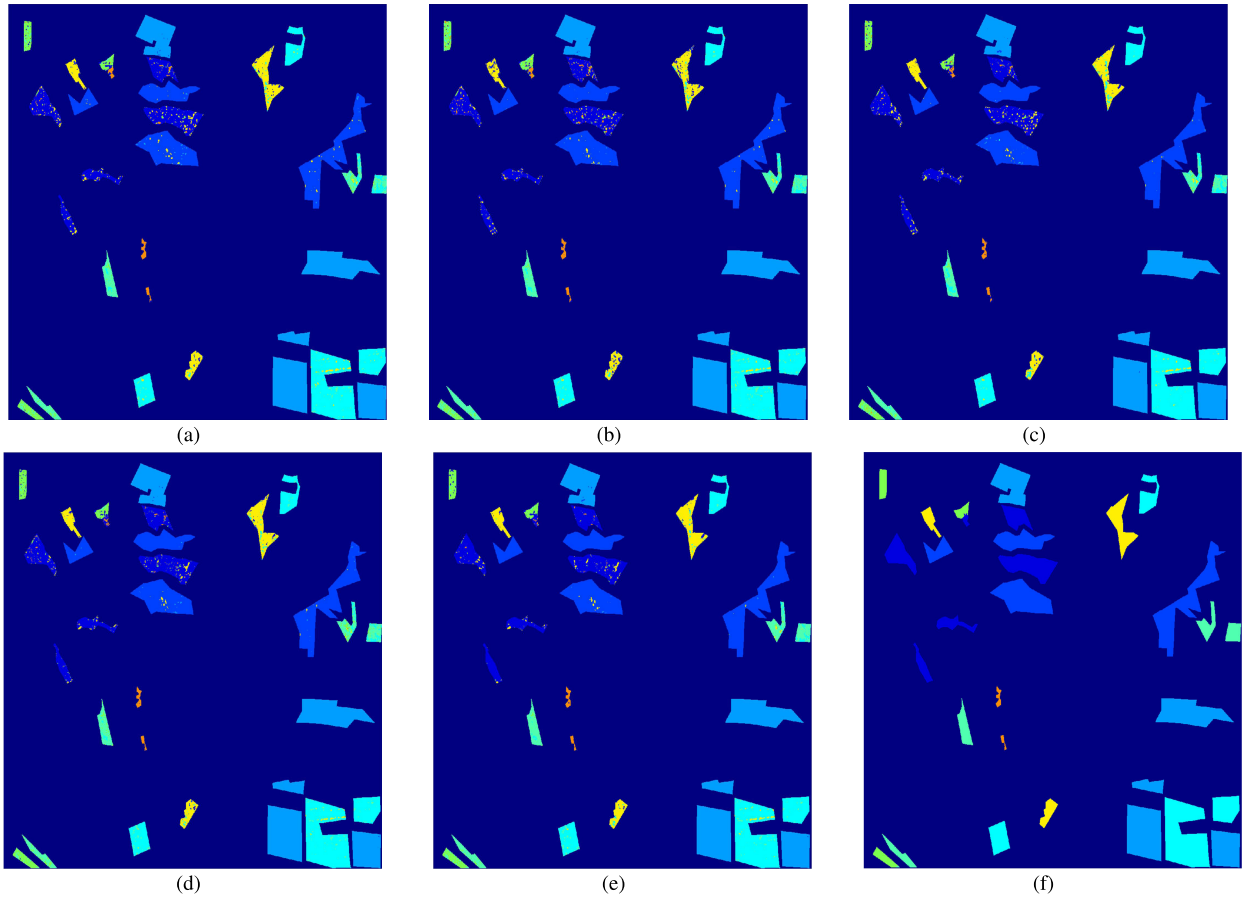


Fig. 12. Classification maps of different methods on the GF-3 data set. (a) CNN-9, OA = 93.69%. (b) CNN, OA = 93.37%. (c) RNPB-CNN, OA = 94.35%. (d) LSTM, OA = 95.05%. (e) RNPB-LSTM, OA = 96.09%. (f) RNPB-LSTM-CRF, OA = 99.64%.

TABLE VIII
CLASSIFICATION ACCURACY (%) COMPARISON ON THE GF-3 DATA SET

Class	CNN -9	CNN	RNPB -CNN	LSTM	RNPB -LSTM	Proposed method
1	96.02	93.72	95.66	95.64	95.34	**98.13**
2	98.96	98.87	99.08	99.36	99.52	**100**
3	99.33	99.83	99.73	99.64	99.95	**100**
4	93.62	94.03	91.75	95.18	96.28	**100**
5	77.02	71.54	85.56	81.35	82.94	**99.99**
6	73.91	78.08	81.32	71.82	89.22	**100**
7	77.44	79.92	79.22	85.66	84.95	**100**
8	48.09	39.04	48.75	59.74	79.09	**100**
OA	93.70	93.37	94.36	95.05	96.09	**99.75**
AA	83.05	81.88	85.13	86.05	90.91	**99.76**
kappa	0.92	0.91	0.92	0.93	0.95	**0.99**

learning performance of 1-D-CNN and LSTM by the synthetic sequence samples; and 3) CRF is a valuable postprocessing tool to further improve the accuracy by the appearance and smoothness kernel of the energy function. On average, these three improvements can increase the OA by approximately 1.7%, 1.6%, and 5.9%, respectively. Though CRF can achieve a significant accuracy improvement, it also has the problems of efficiency degradation, parameter tuning, and algorithm complexity. For the RNPB method, it only yields diverse synthetic samples for network training, regardless of the inference process. In this sense, the method overcomes the three main types of problems in CRF.

D. Classification Efficiency

The Flevoland I data set is chosen as an example to evaluate the classification efficiency of the proposed method, whose computational cost mainly lies in the LSTM network, the RNPB method, and the CRF optimization. As such, CNN and LSTM, as well as their combinations with the RNPB method, are utilized for the comparison purpose. In the experiments, 3188 sequence samples are used for network training, and 157 296 sequence samples are employed for network inference, as shown in Table III. The numbers of network parameters of CNN and LSTM are, respectively, 35 311 and 185 263. The training and inference times of five different methods are shown in Table IX.

Comparing the LSTM network with the CNN network, it can be seen that the LSTM network converges faster in the training phase, and its inference time is slower due to the 185 263 network parameters. Even so, its total running time is still better than the CNN network. After the augmentation of the RNPB method, the training time of the two networks is increased by around four times, which is basically proportional to the increase of the synthetic RNPB samples. On the other hand, since the RNPB method does not affect the inference phase, the inference time remains at the same level. As for the CRF optimization, it will take about 9 s to complete the postprocessing of the probability graph. As far as the inference

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TABLE IX

TIME CONSUMPTION (S) OF DIFFERENT METHODS ON THE 15-CLASS FLEVOLAND I DATA SET

Method	Training Stage	Inference Stage	Total
CNN	201.89	13.06	214.95
LSTM	135.24	40.22	175.46
RNPB-CNN	849.71	13.14	862.85
RNPB-LSTM	606.62	40.67	647.29
RNPB-LSTM-CRF	607.63	48.99	656.62

TABLE X

OVERALL CLASSIFICATION ACCURACY (%) COMPARISON WITH DIFFERENT METHODS ON THE 15-CLASS FLEVOLAND I DATA SET

Method	1% sample rate	0.5% sample rate
CV-CNN [25]	65.18	64.45
SRDNN [40]	94.66	-
SPGraphCNN [61]	98.82	-
RNPB-LSTM	91.25	89.41
RNPB-LSTM-CRF	**99.66**	**98.25**

time is concerned, the proposed method spends more time than the classical CNN network, indicating that the computational complexity of the LSTM network and CRF optimization is high. In future work, more accelerating methods will be studied to improve classification efficiency.

E. Comparison Experiment

To evaluate the classification performance under the condition of limited samples, 1% and 0.5% are set as the training sample rate in the comparison experiment. Three methods are selected for comparison: the complex-valued CNN-based method (CV-CNN) [25], superpixel restrained DNN-based method (SRDNN) [40], and superpixel initialized graph-based CNN (SPGraphCNN) [61]. Similar to the idea of the proposed method, the last two methods are based on deep learning and image segmentation techniques.

Although the network structures of comparison methods are clear, the pretraining process and the parameter fine-tuning will definitely impact the final results. Therefore, the classification results of the 1% sample rate are directly quoted from the corresponding published articles. In the existing methods, there are no discussions about the classification of the 0.5% training sample rate. Since CV-CNN is an open-source PolSAR classification method, it is employed to generate the comparison results with the 0.5% training sample rate.

From Table X, it can be found that the RNPB-LSTM-CRF method can achieve a competitive score under the conditions of a small number of training samples. With a 1% sample rate, this method outperforms the CV-CNN, SRDNN, and SPGraphCNN methods by 34.48%, 5%, and 0.84%, respectively. With the 0.5% sample rate, the classical CV-CNN is improved by 33.75%. On the other hand, under two conditions, CRF improves the OA by 8.41% and 8.84%, respectively. Similarly, the SPGraphCNN method leads to approximately 10% improvement by employing an image segmentation technique. This result indicates that image segmentation techniques can effectively improve the accuracy of deep learning-based small-sample classification.

IV. CONCLUSION

This article proposes a novel classification framework for PolSAR imagery, which integrates the LSTM network with the CRF model for pixel sequence feature learning. The LSTM network tends to mine the spatial polarimetric features among the neighborhood pixels to improve the discrimination. The CRF model is mainly employed to distill the spatial probability information to reduce the intraclass misjudgments. More importantly, a data augmentation method for training samples, namely, the RNPB method, is proposed to improve the small sample learning capability. The experimental results show that the proposed RNPB-based LSTM-CRF method can reach an accuracy of 99.99%, 98.94%, and 99.75% for the three PolSAR data sets, Flevoland I, Flevoland II, and GF-3, respectively. Compared with the classical CNN-based method, the LSTM, RNPB, and CRF methods can independently increase the accuracy. Moreover, this framework with contextual learning and data augmentation is especially suitable for small sample classification because the average training sample ratio can be reduced to only 0.5%. Therefore, this framework offers the potential for practical applications.

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